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REDUCING THE TIME TO SWITCH BETWEEN THE REDUNDANT CLOCK SIGNALS APPLIED TO A PHASE LOCK LOOP

Abstract

A circuit for reducing a time to switch between the redundant clock signals applied to a phase lock loop. The circuit includes a phase frequency detector of the phase lock loop that detects a shift in the phase of a reference clock signal relative to the phase of a feedback clock signal. The circuit further includes a coarse tuning mechanism configured to delay the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal. Furthermore, the circuit includes a fine tuning mechanism configured to align the edge of the feedback clock signal with the edge of the backup reference clock signal, such as by utilizing a series of delay elements to delay the backup reference clock signal at different points in time and selecting the appropriate delayed backup reference clock signal whose edge is aligned with the edge of the feedback clock signal.

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Background/Summary

TECHNICAL FIELD

[0001] The present disclosure relates generally to a phase lock loop, and more particularly to reducing the time to switch between the redundant input clock signals applied to a phase lock loop by utilizing the phase frequency detector of the phase lock loop to sense for a switchover condition.

BACKGROUND

[0002] Phase lock loops (PLL) are found in a myriad of electronic applications, such as communication receivers and clock synchronization circuits in computer systems, for providing a reference signal with a known phase for clocking incoming and out-going data. A conventional PLL includes a phase detector for monitoring the phase difference between an input clock signal and the output signal of a voltage controlled oscillator (VCO) and generating an UP control signal and a DOWN control signal for a charge pump circuit which charges and discharges a loop filter at the input of the VCO. The UP and DOWN control signals drive the VCO to maintain a predetermined phase relationship between signals applied to the phase detector.

[0003] The output signal of the PLL must maintain a predetermined frequency of operation to be useful as a reference for clocking the incoming and out-going data. If the frequency of the input clock signal should drift, or even change to a radically different rate, the output signal of the VCO follows along and attempts to re-achieve phase lock thereto. Such behavior is inherent in the operation of the PLL.

[0004] Currently, phase lock indicators utilize various ways of detecting and reporting a momentary loss of phase lock. Yet, most if not all conventional phase lock indicators cannot distinguish the input frequency. Therefore, the output signal of the VCO locks to the new, albeit incorrect, frequency of the input clock signal, and the lock indicator again reports a valid phase lock status. The reference signal thus clocks the incoming and out-going data at the wrong points resulting in erroneous communication.

[0005] The input clock signal may also become stuck at the logical value of one or stuck at the value of zero causing the PLL to permanently lose phase lock. Since the PLL cannot lock to a DC signal, the phase lock indicator suspends the system operation. While it is informative to know of the permanent loss of phase lock, the phase lock indicator does nothing toward restoring operation of the system which may remain down until the input clock signal is repaired. In many applications, it is desirable and even imperative that the PLL remain operational even if the primary input clock signal becomes invalid as a reference.

[0006] As a result, a design has been introduced involving applying redundant input clocks signals to the phase lock loop. In such a design, the validity of the first input clock signal is determined. If the first input clock signal becomes invalid, then there is a switch to utilize the second input clock signal. Such switching is referred to as a "switchover."

[0007] Unfortunately, such switchovers may occur over a lengthy duration of time (e.g., 156 picoseconds) that results in discontinuities with the interface bit rate thereby causing difficulties in tracking such switchovers.

SUMMARY

[0008] In one embodiment of the present disclosure, a circuit for reducing a time to switch between redundant clock signals applied to a phase lock loop comprises the phase lock loop comprising a phase frequency detector configured to detect a shift in a phase of a reference clock signal of two redundant input clock signals applied to the phase lock loop relative to a phase of a feedback clock signal. The circuit further comprises a coarse tuning mechanism configured to delay the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal of the two redundant input clock signals. The circuit additionally comprises a fine tuning mechanism

configured to align an edge of the feedback clock signal with an edge of the backup reference clock signal of the two redundant input clock signals.

[0009] Additionally, in one embodiment of the present disclosure, the fine tuning mechanism comprises one or more delay elements configured to delay the backup reference clock signal.

[0010] Furthermore, in one embodiment of the present disclosure, the one or more delay elements comprise one or more inverters.

[0011] Additionally, in one embodiment of the present disclosure, the fine tuning mechanism comprises one or more latches.

[0012] Furthermore, in one embodiment of the present disclosure, the one or more latches comprise one or more flip-flops.

[0013] Additionally, in one embodiment of the present disclosure, the fine tuning mechanism comprises an architecture for converting a thermometer code to decimal logic based on outputs of one or more latches.

[0014] Furthermore, in one embodiment of the present disclosure, the architecture comprises a plurality of XOR logic gates

[0015] Additionally, in one embodiment of the present disclosure, an output of the architecture is used for selecting one of a plurality of delayed backup reference clock signals whose edge is aligned with the edge of the feedback clock signal.

[0016] Furthermore, in one embodiment of the present disclosure, the selecting is performed by a multiplexer.

[0017] Additionally, in one embodiment of the present disclosure, the coarse tuning mechanism comprises a shift register or a counter.

[0018] Furthermore, in one embodiment of the present disclosure, the coarse tuning mechanism comprises a divider configured to divide a frequency of the feedback clock signal to match a frequency of the backup reference clock signal.

[0019] In another embodiment of the present disclosure, a method for reducing a time to switch between redundant clock signals applied to a phase lock loop comprises detecting a shift in a phase of a reference clock signal of two redundant input clock signals applied to the phase lock loop relative to a phase of a feedback clock signal. The method further comprises performing a coarse tuning operation corresponding to delaying the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal of the two redundant input clock signals. The method additionally comprises performing a fine tuning operation corresponding to aligning an edge of the feedback clock signal with an edge of the backup reference clock signal.

[0020] Furthermore, in one embodiment of the present disclosure, the method additionally comprises dividing a frequency of the feedback clock signal to match a frequency of the backup reference clock signal by a divider.

[0021] Additionally, in one embodiment of the present disclosure, the method further comprises setting the divider to a period of a clock frequency.

[0022] Furthermore, in one embodiment of the present disclosure, the method additionally comprises outputting the feedback clock signal by a selector in response to detecting the shift in the phase of the reference clock signal relative to the phase of the feedback clock signal thereby causing a phase frequency detector of the phase lock loop to receive the feedback clock signal at both inputs of the phase frequency detector.

[0023] Additionally, in one embodiment of the present disclosure, a time duration of the aligning of the edge of the feedback clock signal with the edge of the backup reference clock signal corresponds to a delay of a delay element.

[0024] Furthermore, in one embodiment of the present disclosure, the delay element comprises an inverter.

[0025] Additionally, in one embodiment of the present disclosure, the method further comprises enabling the feedback clock signal to latch data into flip-flops, where the data corresponds to an

edge of one of a plurality of delayed backup reference clock signals.

[0026] Furthermore, in one embodiment of the present disclosure, the method additionally comprises converting a thermometer code to decimal logic based on outputs of the flip-flops. Furthermore, the method comprises selecting one of the plurality of delayed backup reference clock signals using the decimal logic whose edge aligns with the edge of the feedback clock signal.

[0027] Additionally, in one embodiment of the present disclosure, the method further comprises propagating the aligned backup reference clock signal to the phase lock loop.

[0028] Accordingly, embodiments of the present disclosure reduce the time duration of the switchover. That is, embodiments of the present disclosure reduce the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop.

[0029] The foregoing has outlined rather generally the features and technical advantages of one or more embodiments of the present disclosure in order that the detailed description of the present disclosure that follows may be better understood. Additional features and advantages of the present disclosure will be described hereinafter which may form the subject of the claims of the present disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0030] A better understanding of the present disclosure can be obtained when the following detailed description is considered in conjunction with the following drawings, in which:

[0031] FIG. 1 illustrates an embodiment of the present disclosure of a circuit for reducing a time to switch between the redundant clock signals applied to a phase lock loop;

[0032] FIG. 2 illustrates the internal circuitry of the phase lock loop, the coarse tuning mechanism and the fine tuning mechanism in accordance with an embodiment of the present invention;

[0033] FIG. 3 is a flowchart of a method for reducing the time to switch between the redundant clock signals applied to a phase lock loop in accordance with an embodiment of the present disclosure;

[0034] FIG. 4 is a flowchart of a method for performing the coarse turning operation in accordance with an embodiment of the present disclosure;

[0035] FIG. 5 is a flowchart of a method for performing the fine tuning operation in accordance with an embodiment of the present disclosure; and

[0036] FIG. 6 is a timing diagram of the signals as a function of time during the switchover case in accordance with an embodiment of the present disclosure.

DETAILED DESCRIPTION

[0037] In one embodiment of the present disclosure, a circuit for reducing a time to switch between redundant clock signals applied to a phase lock loop comprises the phase lock loop comprising a phase frequency detector configured to detect a shift in a phase of a reference clock signal of two redundant input clock signals applied to the phase lock loop relative to a phase of a feedback clock signal. The circuit further comprises a coarse tuning mechanism configured to delay the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal of the two redundant input clock signals. The circuit additionally comprises a fine tuning mechanism configured to align an edge of the feedback clock signal with an edge of the backup reference clock signal of the two redundant input clock signals.

[0038] In this manner, the time duration of the switchover is greatly reduced. That is, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop is greatly reduced.

[0039] Additionally, in one embodiment of the present disclosure, the fine tuning mechanism

comprises one or more delay elements configured to delay the backup reference clock signal.

[0040] In this manner, the edge of the backup reference clock signal can be aligned with the edge of the feedback clock signal.

[0041] Furthermore, in one embodiment of the present disclosure, the one or more delay elements comprise one or more inverters.

[0042] In this manner, the duration of the switchover can correspond to the delay of a single delay element, such as an inverter.

[0043] Additionally, in one embodiment of the present disclosure, the fine tuning mechanism comprises one or more latches.

[0044] In this manner, the data corresponding to an edge of the backup reference clock signal is latched into the latches.

[0045] Furthermore, in one embodiment of the present disclosure, the one or more latches comprise one or more flip-flops.

[0046] In this manner, the data corresponding to an edge of the backup reference clock signal is latched into the one or more flip-flops.

[0047] Additionally, in one embodiment of the present disclosure, the fine tuning mechanism comprises an architecture for converting a thermometer code to decimal logic based on outputs of one or more latches.

[0048] In this manner, the appropriate delayed backup reference clock signal is selected based on the decimal logic.

[0049] Furthermore, in one embodiment of the present disclosure, the architecture comprises a plurality of XOR logic gates

[0050] In this manner, the thermometer code is converted to decimal logic based on the outputs of the latch(es).

[0051] Additionally, in one embodiment of the present disclosure, an output of the architecture is used for selecting one of a plurality of delayed backup reference clock signals whose edge is aligned with the edge of the feedback clock signal.

[0052] In this manner, the appropriate delayed backup reference clock signal is selected corresponding to the delayed backup reference clock signal whose edge aligns with the edge of the feedback clock signal.

[0053] Furthermore, in one embodiment of the present disclosure, the selecting is performed by a multiplexer.

[0054] In this manner, a multiplexer is used for outputting the delayed backup reference clock signal whose edge aligns with the edge of the feedback clock signal.

[0055] Additionally, in one embodiment of the present disclosure, the coarse tuning mechanism comprises a shift register or a counter.

[0056] In this manner, the frequency of the feedback clock signal can be appropriately divided so as to match the frequency of the backup reference clock signal.

[0057] Furthermore, in one embodiment of the present disclosure, the coarse tuning mechanism comprises a divider configured to divide a frequency of the feedback clock signal to match a frequency of the backup reference clock signal.

[0058] In this manner, coarse tuning is performed so as to enable the switchover to occur with a period of the clock frequency.

[0059] In another embodiment of the present disclosure, a method for reducing a time to switch between redundant clock signals applied to a phase lock loop comprises detecting a shift in a phase of a reference clock signal of two redundant input clock signals applied to the phase lock loop relative to a phase of a feedback clock signal. The method further comprises performing a coarse tuning operation corresponding to delaying the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal of the two redundant input clock signals. The method additionally comprises performing a fine tuning operation corresponding to aligning an

edge of the feedback clock signal with an edge of the backup reference clock signal.

[0060] In this manner, the time duration of the switchover is greatly reduced. That is, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop is greatly reduced.

[0061] Furthermore, in one embodiment of the present disclosure, the method additionally comprises dividing a frequency of the feedback clock signal to match a frequency of the backup reference clock signal by a divider.

[0062] In this manner, coarse tuning is performed so as to enable the switchover to occur with a period of the clock frequency.

[0063] Additionally, in one embodiment of the present disclosure, the method further comprises setting the divider to a period of a clock frequency.

[0064] In this manner, coarse tuning is performed so as to enable the switchover to occur with a period of the clock frequency.

[0065] Furthermore, in one embodiment of the present disclosure, the method additionally comprises outputting the feedback clock signal by a selector in response to detecting the shift in the phase of the reference clock signal relative to the phase of the feedback clock signal thereby causing a phase frequency detector of the phase lock loop to receive the feedback clock signal at both inputs of the phase frequency detector.

[0066] In this manner, the alignment cycle during the performance of fine tuning can occur.

[0067] Additionally, in one embodiment of the present disclosure, a time duration of the aligning of the edge of the feedback clock signal with the edge of the backup reference clock signal corresponds to a delay of a delay element.

[0068] In this manner, the duration of the switchover can correspond to the delay of a single delay element.

[0069] Furthermore, in one embodiment of the present disclosure, the delay element comprises an inverter.

[0070] In this manner, the duration of the switchover can correspond to the delay of a single delay element, such as an inverter.

[0071] Additionally, in one embodiment of the present disclosure, the method further comprises enabling the feedback clock signal to latch data into flip-flops, wherein the data corresponds to an edge of one of a plurality of delayed backup reference clock signals.

[0072] In this manner, the data corresponding to one of the delayed backup reference clock signals is latched into the one or more flip-flops.

[0073] Furthermore, in one embodiment of the present disclosure, the method additionally comprises converting a thermometer code to decimal logic based on outputs of said flip-flops. Furthermore, the method comprises selecting one of the plurality of delayed backup reference clock signals using the decimal logic whose edge aligns with the edge of the feedback clock signal.

[0074] In this manner, the appropriate delayed backup reference clock signal is selected based on the decimal logic.

[0075] Additionally, in one embodiment of the present disclosure, the method further comprises propagating the aligned backup reference clock signal to the phase lock loop.

[0076] In this manner, the phase lock loop will lock onto the appropriate reference clock signal.

[0077] As stated above, phase lock loops (PLL) are found in a myriad of electronic applications, such as communication receivers and clock synchronization circuits in computer systems, for providing a reference signal with a known phase for clocking incoming and out-going data. A conventional PLL includes a phase detector for monitoring the phase difference between an input clock signal and the output signal of a voltage controlled oscillator (VCO) and generating an UP control signal and a DOWN control signal for a charge pump circuit which charges and discharges a loop filter at the input of the VCO. The UP and DOWN control signals drive the VCO to maintain a predetermined phase relationship between signals applied to the phase detector.

[0078] The output signal of the PLL must maintain a predetermined frequency of operation to be useful as a reference for clocking the incoming and out-going data. If the frequency of the input clock signal should drift, or even change to a radically different rate, the output signal of the VCO follows along and attempts to re-achieve phase lock thereto. Such behavior is inherent in the operation of the PLL.

[0079] Currently, phase lock indicators utilize various ways of detecting and reporting a momentary loss of phase lock. Yet, most if not all conventional phase lock indicators cannot distinguish the input frequency. Therefore, the output signal of the VCO locks to the new, albeit incorrect, frequency of the input clock signal, and the lock indicator again reports a valid phase lock status. The reference signal thus clocks the incoming and out-going data at the wrong points resulting in erroneous communication.

[0080] The input clock signal may also become stuck at the logical value of one or stuck at the value of zero causing the PLL to permanently lose phase lock. Since the PLL cannot lock to a DC signal, the phase lock indicator suspends the system operation. While it is informative to know of the permanent loss of phase lock, the phase lock indicator does nothing toward restoring operation of the system which may remain down until the input clock signal is repaired. In many applications, it is desirable and even imperative that the PLL remain operational even if the primary input clock signal becomes invalid as a reference.

[0081] As a result, a design has been introduced involving applying redundant input clocks signals to the phase lock loop. In such a design, the validity of the first input clock signal is determined. If the first input clock signal becomes invalid, then there is a switch to utilize the second input clock signal. Such switching is referred to as a "switchover."

[0082] Unfortunately, such switchovers may occur over a lengthy duration of time (e.g., 156 picoseconds) that results in discontinuities with the interface bit rate thereby causing difficulties in tracking such switchovers.

[0083] The embodiments of the present disclosure provide a means for reducing the duration of time of the switchover by performing coarse tuning and fine tuning. In one embodiment, coarse tuning is utilized for aligning the feedback clock signal, which is used to detect a shift in the phase of a reference clock signal (one of the redundant input clock signals to the phase lock loop), with the backup reference clock signal (the other redundant input clock signal to the phase lock loop). In one embodiment, such coarse tuning is performed by dividing the frequency of the feedback clock signal by a divider to match the frequency of the backup reference clock signal. Such a coarse tuning enables the switchover to occur with a period of the clock frequency, where the clock frequency signal matches the frequency of the backup reference clock signal by enabling an N-1 mode within the divider. The N-1 mode is then enabled by the redundant reference clock signal. Such a coarse tuning enables the switchover to occur with a period of the pre-divided clock frequency. In one embodiment, fine tuning is utilized for aligning the edge of the feedback clock signal with the edge of the backup reference clock signal. In one embodiment, fine tuning is performed by utilizing a series of delay elements to delay the backup reference clock signal at different points in time. In one embodiment, fine tuning is performed by enabling the feedback clock signal to latch data into latches, such as flip-flops, where the data corresponds to an edge of one of the delayed backup reference clock signals. The outputs of such latches are used by an architecture (e.g., XOR logic gates) to convert a thermometer code to decimal logic based on such outputs which is used by a selector to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge is aligned with the edge of the feedback clock signal. In one embodiment, such an alignment is performed within the duration of time of a single delay element. In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop is greatly reduced. A further description of these and other features will be provided below.

[0084] In some embodiments of the present disclosure, the present disclosure comprises a circuit

for reducing a time to switch between the redundant clock signals applied to a phase lock loop. In one embodiment of the present disclosure, a phase frequency detector of the phase lock loop of the circuit detects a shift in the phase of a reference clock signal (one of the redundant input clock signals applied to the phase lock loop) relative to the phase of a feedback clock signal. A feedback clock signal, as used herein, refers to the clock signal outputted by the phase lock loop. The circuit further includes a coarse tuning mechanism configured to perform coarse tuning involving delaying the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal (the other redundant input clock signal) in response to detecting the shift in the phase of the reference clock signal relative to the phase of the feedback clock signal. In one embodiment, such coarse tuning is performed by dividing the frequency of the feedback clock signal by a divider to match the frequency of the backup reference clock signal. Such a coarse tuning enables the switchover to occur with a period of the pre-divided clock frequency. Furthermore, the circuit includes a fine tuning mechanism configured to perform fine tuning involving aligning the edge of the feedback clock signal with the edge of the backup reference clock signal. In one embodiment, fine tuning is performed by utilizing a series of delay elements to delay the backup reference clock signal at different points in time. In one embodiment, fine tuning is performed by enabling the feedback clock signal to latch data into latches, such as flip-flops, where the data corresponds to an edge of one of the delayed backup reference clock signals. The outputs of such latches are used by an architecture (e.g., XOR logic gates) to convert a thermometer code to decimal logic based on such outputs which is used by a selector to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge is aligned with the edge of the feedback clock signal. In one embodiment, such an alignment is performed within the duration of time of a single delay element. As a result, the switchover occurs within the duration of time (e.g., 19 ps) of a single delay element, such as an inverter. In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop is greatly reduced.

[0085] In the following description, numerous specific details are set forth to provide a thorough understanding of the present disclosure. However, it will be apparent to those skilled in the art that the present disclosure may be practiced without such specific details. In other instances, well-known circuits have been shown in block diagram form in order not to obscure the present disclosure in unnecessary detail. For the most part, details considering timing considerations and the like have been omitted inasmuch as such details are not necessary to obtain a complete understanding of the present disclosure and are within the skills of persons of ordinary skill in the relevant art.

[0086] Referring now to the Figures in detail, FIG. 1 illustrates an embodiment of the present disclosure of a circuit **100** for reducing a time to switch between the redundant clock signals applied to a phase lock loop.

[0087] As shown in FIG. 1, circuit **100** includes a phase lock loop (PLL) **101** configured to provide a signal (“feedback clock signal”) with a known phase for clocking incoming and out-going data.

[0088] In one embodiment, PLL **101** is configured to receive a “reference clock signal” **104** (“REFCLK”). In one embodiment, reference clock signal **104** corresponds to one of a pair of redundant reference clock signals. In one embodiment, PLL **101** is configured to detect a shift in the phase of reference clock signal **104** relative to the phase of a “feedback clock signal” **105**. Feedback clock signal **105** (“FBCLK”), as used herein, refers to the clock signal outputted by PLL **101**.

[0089] In one embodiment, upon detecting a shift in reference clock signal **104** relative to the phase of feedback clock signal **105**, coarse tuning is performed by coarse tuning mechanism **102**. “Coarse tuning,” as used herein, refers to delaying feedback clock signal **105** until feedback clock signal **105** is aligned with the backup reference clock signal. The “backup reference clock signal,” as used herein, refers to the other reference clock signal of the pair of redundant reference clock signals

that was not identified as having its phase shifted relative to the phase of feedback clock signal **105**. In one embodiment, coarse tuning mechanism **102** performs coarse tuning, i.e., aligns the feedback clock signal with the backup reference clock signal, by delaying the feedback clock signal until the feedback clock signal is aligned with the backup reference clock signal, where the backup reference clock signal utilizes an N-1 mode.

[0090] In one embodiment, such an aligned feedback clock signal is then inputted to PLL **101** along with reference clock signal **104** ("REFCLK"), which in the case of detecting the shift in the phase of reference clock signal **104** relative to the phase of feedback clock signal **105** corresponds to a clock signal matching feedback clock signal **105** during a period of time, referred to herein as the "blackout period," as discussed further below.

[0091] Furthermore, in one embodiment, upon detecting a shift in reference clock signal **104** relative to the phase of feedback clock signal **105**, fine tuning is performed by fine tuning mechanism **103**.

[0092] "Fine tuning," as used herein, refers to aligning the edge of the feedback clock signal with the edge of the backup reference clock signal.

[0093] In one embodiment, fine tuning is performed by utilizing a series of delay elements to delay the backup reference clock signal at different points in time. In one embodiment, fine tuning is performed by enabling the feedback clock signal to latch data into latches, such as flip-flops, where the data corresponds to an edge of one of the delayed backup reference clock signals. The outputs of such latches are used by an architecture (e.g., XOR logic gates) to convert a thermometer code to decimal logic based on such outputs which is used by a selector to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge is aligned with the edge of the feedback clock signal. In one embodiment, such an alignment is performed within the duration of time of a single delay element. In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop (e.g., PLL **101**) is greatly reduced.

[0094] A further description of such features, including the internal circuitry of PLL **101**, coarse tuning mechanism **102** and fine tuning mechanism **103** is provided below in connection with FIG. 2.

[0095] FIG. 2 illustrates the internal circuitry of the phase lock loop (e.g., PLL **101**), the coarse tuning mechanism (e.g., coarse tuning mechanism **102**) and the fine tuning mechanism (e.g., fine tuning mechanism **103**) in accordance with an embodiment of the present invention.

[0096] As shown in FIG. 2, PLL **101** includes a phase frequency detector (PFD) **201** which compares the phase between two input signals, such as reference clock signal **104** and feedback clock signal **105**, and generates outputs based on the phase difference between them. In one embodiment, PFD **201** corresponds to an edge triggered JK flip-flop phase frequency detector. In one embodiment, PFD **201** corresponds a dual D type phase comparator. In one embodiment, feedback clock signal **105** is the output signal of a voltage controlled oscillator **204**.

[0097] In one embodiment, upon detecting a phase difference between reference clock signal **104** and feedback clock signal **105**, PFD **201** generates an UP control signal and a DOWN control signal for a charge pump (CP) **203** which charges and discharges a loop filter **202** at the input of VCO **204**. The UP and DOWN control signals drive VCO **204** to maintain a predetermined phase relationship between the signals applied to PFD **201**.

[0098] The output signal of PLL **101** must maintain a predetermined frequency of operation to be useful as a reference for clocking the incoming and out-going data. If the frequency of the input clock signal should drift, or even change to a radically different rate, the output signal of VCO **204** follows along and attempts to re-achieve phase lock thereto. Such behavior is inherent in the operation of PLL **101**.

[0099] Furthermore, as shown in FIG. 2, coarse tuning mechanism **102** includes a divider ("FBDIV") **205** configured to divide a frequency of feedback clock signal **105** from PLL **101** (e.g.,

6.4 GHz) to match the frequency of the backup reference clock signal. A divider, such as divider **205**, as used herein, refers to a counter. In one embodiment, the functionality of divider **205** is performed by a shift register. In one embodiment, such a divider **205** is digitally programmable. As will be discussed in greater detail below, upon detecting a phase difference between reference clock signal **104** and feedback clock signal **105**, the backup reference clock signal is selected to replace reference clock signal **104** whose phase does not match the phase of feedback clock signal **105**. The “backup reference clock signal,” as used herein, refers to the other reference clock signal of the pair of redundant reference clock signals that was not identified as having its phase shifted relative to the phase of feedback clock signal **105**.

[0100] In one embodiment, after divider **205** divides the frequency of feedback clock signal **105** from PLL **101** (e.g., 6.4 GHz) to match the frequency of the backup reference clock signal, divider **205** is set to a period of the clock frequency (e.g., 156 ps based on a clock frequency of 6.4 GHz) based on setting the value of the control logic signal, FBDIV_FREEZE signal **206**, to the logical value of one.

[0101] In one embodiment, coarse tuning mechanism **102** further includes a selector **207** (“FBSEL”) configured to select either feedback clock signal **105** or reference clock signal **104** emanating from fine tuning mechanism **103**.

[0102] In one embodiment, as discussed in greater detail below, the output of selector **207** consists of reference clock signal **104** that corresponds to feedback clock signal **105**, such as in the situation after coarse tuning is performed. In such a situation, PFD **201** receives feedback clock signal **105** at both its inputs. Such a situation occurs during a period of time referred to herein as the “blackout period” as discussed further below.

[0103] Additionally, as shown in FIG. 2, fine tuning mechanism **103** includes the redundant input clock signals to PLL **101**, such as reference clock signal A **208A** (“REFA”) and reference clock signal B **208B** (“REFB”). One of such reference clock signals **208A**, **208B** corresponds to the reference clock signal (REFCLK) **104** that is selected to be inputted to PLL **101**.

[0104] In one embodiment, such signals **208A**, **208B** are inputted to a selector (“REFSEL”) **209** configured to select one of such reference clock signals **208A**, **208B** (identified as “PRE_REFCLK” **210** in FIG. 2).

[0105] As will be discussed in greater detail below, such a reference clock signal **210** (“PRE_REFCLK”) is delayed at various time intervals by a series of delay elements, such as inverter logic gates **211A-211N**. Inverter logic gates **211A-211N** may collectively or individually be referred to as inverters **211** or inverter **211**, respectively.

[0106] In one embodiment, upon initiating fine tuning after detecting a shift in the phase of reference clock signal **104** with respect to the phase of feedback clock signal **105**, selector **209** selects the backup reference clock signal. In such an embodiment, PRE-REFCLK **210** corresponds to the backup reference clock signal. For example, if REFA **208A** corresponds to reference clock signal **104** whose phase was shifted with respect to the phase of feedback clock signal **105**, then the backup reference clock signal would be REFB **208B**.

[0107] Furthermore, as shown in FIG. 2, fine tuning mechanism **103** includes a logic gate, such as an AND logic gate **212**, that receives feedback clock signal **105** as well as control signal (“FINE_CAL_EN”) **213** configured to enable fine tuning to be performed upon being set to the logical value of one. In one embodiment, such a control signal enables fine tuning to be performed at the time divider **205** is set to the period of the clock frequency.

[0108] In one embodiment, the output of AND logic gate **212** enables feedback clock signal **105** to latch data into a series of flip-flops **214A-214N**, where the data corresponds to an edge of one of the delayed backup reference clock signals. Flip-flops **214A-214N** may collectively or individually be referred to as flip-flops **214** or flip-flop **214**, respectively.

[0109] In one embodiment, flip-flops **214** receive a series of delayed backup reference clock signals (identified as X.sub.0 . . . X.sub.15). Such delayed backup reference signals correspond to

the reference clock signal (e.g., REFB **208B**) selected by selector **209** that was delayed by inverters **211** at different time intervals. The outputs of flip-flops **214** correspond to a series of values (e.g., X2.sub.0 . . . X2.sub.15) which are used by an architecture, such as XOR logic gates **215A-215N**, to convert a thermometer code to decimal logic based on such outputs which is used by a selector **216** (e.g., multiplexer) to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge aligns with the edge of feedback clock signal **105**. XOR logic gates **215A-215N** may collectively or individually be referred to as XOR logic gates **215** or XOR logic gate **215**, respectively. A “thermometer code,” as used herein, refers to representing a value by a sequence of 0s followed by a sequence of 1s, where the number of 1s corresponds to the value to be represented by the thermometer code. “Decimal logic,” as used herein, refers to a value (base-10 representation) that is used to represent one of the delayed backup reference clock signals (e.g., one of the 16 delayed backup reference clock signals). In one embodiment, the decimal logic outputted by XOR logic gates **215** is used as the control signal **217** (“FINE_SEL”) to enable selector **216** (e.g., multiplexer) to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge aligns with the edge of feedback clock signal **105**.

[0110] In one embodiment, such an alignment is performed within the duration of time (e.g., 19 ps) of a single delay element, such as an inverter (e.g., inverter **211**). In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop (e.g., PLL **101**) is greatly reduced.

[0111] A further discussion regarding the functionality of the components of circuit **100** is provided below in connection with FIGS. 3-6. FIG. 3 is a flowchart of a method for reducing the time to switch between the redundant clock signals applied to a phase lock loop. FIG. 4 is a flowchart of a method for performing the coarse tuning operation. FIG. 5 is a flowchart of a method for performing the fine tuning operation. FIG. 6 is a timing diagram of the signals as a function of time during the switchover case.

[0112] As stated above, FIG. 3 is a flowchart of a method **300** for reducing the time to switch between the redundant clock signals applied to a phase lock loop (e.g., PLL **101**) in accordance with an embodiment of the present disclosure.

[0113] Referring to FIG. 3, in conjunction with FIGS. 1-2, in step **301**, PFD **201** of PLL **101** detects a shift in the phase of reference clock signal **104** (REFCLK) relative to a phase of feedback clock signal **105** (FBCLK). As discussed above, such a reference clock signal is one of the redundant input clock signals **208** applied to PLL **101**, such as REFA **208A**.

[0114] Furthermore, as stated above, in one embodiment, PFD **201** corresponds to an edge triggered JK flip-flop phase frequency detector. In one embodiment, PFD **201** corresponds a dual D type phase comparator.

[0115] In step **302**, coarse tuning mechanism **102** performs a coarse tuning operation. As discussed above, “coarse tuning,” as used herein, refers to delaying feedback clock signal **105** until feedback clock signal **105** is aligned with the backup reference clock signal. The “backup reference clock signal,” as used herein, refers to the other reference clock signal (e.g., REFB **208B**) of the pair of redundant reference clock signals **208** that was not identified as having its phase shifted relative to the phase of feedback clock signal **105**.

[0116] A discussion regarding performing coarse tuning is provided below in connection with FIG. 4.

[0117] FIG. 4 is a flowchart of a method **400** for performing the coarse tuning operation in accordance with an embodiment of the present disclosure.

[0118] Referring to FIG. 4, in conjunction with FIGS. 1-3, in step **401**, divider **205** (“FBDIV”) divides a frequency of feedback clock signal **105** from PLL **101** to match the frequency of the backup reference clock signal. In this manner, feedback clock signal **105** is delayed until feedback clock signal **105** is aligned with the backup reference clock signal.

[0119] As stated above, in one embodiment, divider **205**, as used herein, refers to a counter. In one embodiment, the functionality of divider **205** is performed by a shift register. In one embodiment, such a divider **205** is digitally programmable.

[0120] In one embodiment, the aligned feedback clock signal discussed above is then inputted to PLL **101** along with reference clock signal **104** (“REFCLK”), which in the case of detecting a shift in reference clock signal **104** relative to the phase of feedback clock signal **105**, corresponds to a signal matching feedback clock signal **105** during a period of time, referred to herein as the “blackout period.”

[0121] That is, during the blackout period, in step **402**, selector **207** outputs reference clock signal **104** that corresponds to feedback clock signal **105** to phase frequency detector **201** of phase lock loop **101** thereby causing phase frequency detector **201** of phase lock loop **101** to receive feedback clock signal **105** at both inputs. During such a blackout period, the alignment cycle during fine tuning occurs as discussed below.

[0122] In step **403**, after divider **205** divides the frequency of feedback clock signal **105** from PLL **101** (e.g., 6.4 GHz) to match the frequency of the backup reference clock signal, divider **205** is set to a period of the clock frequency (e.g., 156 ps based on a clock frequency of 6.4 GHz) based on setting the value of the control logic signal, FBDIV_FREEZE signal **206**, to the logical value of one.

[0123] Returning to FIG. **3**, in conjunction with FIGS. **1-2** and **4**, in step **303**, fine tuning mechanism **103** performs a fine tuning operation. “Fine tuning,” as used herein, refers to aligning the edge of the feedback clock signal (e.g., feedback clock signal **105**) with the edge of the backup reference clock signal.

[0124] A discussion regarding performing fine tuning is provided below in connection with FIG. **5**.

[0125] FIG. **5** is a flowchart of a method **500** for performing the fine tuning operation in accordance with an embodiment of the present disclosure.

[0126] Referring to FIG. **5**, in conjunction with FIGS. **1-4**, in step **501**, upon PFD **201** of PLL **101** detecting a shift in the phase of reference clock signal **104** (REFCLK) relative to a phase of feedback clock signal **105** (FBCLK), selector **209** selects the backup reference clock signal (e.g., reference clock signal **208B**) so as to switch over to use the backup reference clock signal as input to PLL **101**. In such an embodiment, PRE-REFCLK **210** corresponds to the backup reference clock signal (e.g., reference clock signal **208B**). As discussed above, the “backup reference clock signal” refers to the other redundant clock signal **208** that was not identified as having its phase shifted relative to the phase of feedback clock signal **105**. For example, if reference clock signal **208A** was identified as having its phase shifted relative to the phase of feedback clock signal **105**, then reference clock signal **208B** is selected as the backup reference clock signal.

[0127] In step **502**, AND logic gate **212** enables feedback clock signal **105** to latch data into a series of flip-flops **214**, where the data corresponds to an edge of one of the delayed backup reference clock signals.

[0128] As stated above, reference clock signal **210** (“PRE_REFCLK”), which corresponds to the backup reference clock signal, is delayed at various time intervals by a series of delay elements, such as inverter logic gates **211**.

[0129] Furthermore, as discussed above, in one embodiment, AND logic gate **212** receives feedback clock signal **105** as well as control signal **213** (“FINE_CAL_EN”) configured to enable fine tuning to be performed upon being set to the logical value of one. In one embodiment, such a control signal enables fine tuning to be performed at the time divider **205** is set to the period of the clock frequency. In particular, such a control signal enables AND logic gate **212** to enable feedback clock signal **105** to latch data into a series of flip-flops **214**, where the data corresponds to an edge of one of the delayed backup reference clock signals.

[0130] In step **503**, XOR logic gates **215** convert a thermometer code to decimal logic based on the outputs of flip-flops **214**.

[0131] As discussed above, in one embodiment, flip-flops **214** receive a series of delayed backup reference clock signals (identified as X.sub.0 . . . X.sub.15). Such delayed backup reference signals correspond to the reference clock signal selected by selector **209** that was delayed by inverters **211** at different time intervals. The outputs of flip-flops **214** correspond to a series of values (e.g., X2.sub.0 . . . X2.sub.15) which are used by an architecture, such as XOR logic gates **215**, to convert a thermometer code to decimal logic based on such outputs which is used by a selector **216** (e.g., multiplexer) to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge aligns with the edge of feedback clock signal **105**. A “thermometer code,” as used herein, refers to representing a value by a sequence of 0s followed by a sequence of 1s, where the number of 1s corresponds to the value to be represented by the thermometer code. “Decimal logic,” as used herein, refers to a value (base-10 representation) that is used to represent one of the delayed backup reference clock signals (e.g., one of the 16 delayed backup reference clock signals).

[0132] In step **504**, selector **216** selects one of the delayed backup reference clock signals using the decimal logic, which corresponds to the backup reference clock signal whose edge aligns with the edge of feedback clock signal **105**.

[0133] As stated above, the decimal logic outputted by XOR logic gates **215** is used as the control signal **217** (“FINE_SEL”) to enable selector **216** (e.g., multiplexer) to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge aligns with the edge of feedback clock signal **105**.

[0134] In one embodiment, the duration of time for such an alignment occurs within the duration of time (e.g., 19 ps) of a single delay element, such as inverter **211**, which corresponds to the time to switch from one redundant clock signal to the other redundant clock signal to be used to be applied to PLL **101**. In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal is greatly reduced.

[0135] In step **505**, the aligned backup reference clock signal is propagated to phase lock loop **101**, such as by selector **216** outputting the aligned backup reference clock signal to the input of selector **207**.

[0136] A timing diagram of the switchover case using the principles of the present disclosure is provided below in connection with FIG. **6**.

[0137] FIG. **6** is a timing diagram of the signals as a function of time during the switchover case.

[0138] Referring to FIG. **6**, in conjunction with FIGS. **1-5**, timing diagram **600** illustrates the values of the signals during the switchover case using the principles of the present disclosure. As shown in FIG. **6**, at point **601**, there was expected to be an edge for REFA **208A**. However, since there was no edge, a phase shift was detected at point **601**, which triggers a shift detected at point **602**, which triggers the fbdiv_freeze signal **206** to be enabled at point **603**, which causes feedback clock signal **105** (FBCLK) to have a value of 0 at point **604**.

[0139] At the next reference clock signal cycle, which is shown by pre-refclock **210** and refclk **104** at point **605**, the reference clock signal (refclk **104**) goes high which releases fbdiv_freeze signal **206** at point **606**, which causes feedback clock signal **105** to go high at point **607**.

[0140] When feedback clock signal **105** goes high, it causes the reference delay line select (ref-dly-line-sel), referring to the architecture of flip-flops **214** and XOR gates **215** in selecting the appropriate delayed backup reference clock signal whose edge is aligned with the edge of feedback clock signal **105**, to latch onto the data (e.g., value of 6) at point **608**. The delay of latching onto the data affects the delay of selector **216** to output the aligned backup reference clock signal (aligned with feedback clock signal **105**), where such a delay (see element **609**) corresponds to the duration of the switchover (duration of time from switching to the other redundant input clock signal **208**), which corresponds to the delay of a delay element (e.g., 19 ps), such as inverter **211**. In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop (e.g., PLL **101**) is greatly reduced.

[0141] Furthermore, as shown in timing diagram **600**, after aligning the backup reference clock signal with feedback clock signal **105**, reference clock signal **104** (refclk) is aligned with feedback clock signal **105** (fbclk) as shown at points **610**, **611**, respectively. Such clock signals are also aligned at the next rising edge at points **612**, **613**, respectively. The aligned reference clock signal **104** (refclk) is then propagated to PLL **101** by selector **207** (fb sel) at point **614**, which corresponds to the end of the blackout period when selector **207** (fb sel) no longer outputs feedback clock signal (fbclk) **105**.

[0142] In this manner, the time duration of the switchover is greatly reduced. That is, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop is greatly reduced.

[0143] Furthermore, the principles of the present disclosure improve the technology or technical field involving a phase lock loop.

[0144] As discussed above, phase lock loops (PLL) are found in a myriad of electronic applications, such as communication receivers and clock synchronization circuits in computer systems, for providing a reference signal with a known phase for clocking incoming and out-going data. A conventional PLL includes a phase detector for monitoring the phase difference between an input clock signal and the output signal of a voltage controlled oscillator (VCO) and generating an UP control signal and a DOWN control signal for a charge pump circuit which charges and discharges a loop filter at the input of the VCO. The UP and DOWN control signals drive the VCO to maintain a predetermined phase relationship between signals applied to the phase detector. The output signal of the PLL must maintain a predetermined frequency of operation to be useful as a reference for clocking the incoming and out-going data. If the frequency of the input clock signal should drift, or even change to a radically different rate, the output signal of the VCO follows along and attempts to re-achieve phase lock thereto. Such behavior is inherent in the operation of the PLL. Currently, phase lock indicators utilize various ways of detecting and reporting a momentary loss of phase lock. Yet, most if not all conventional phase lock indicators cannot distinguish the input frequency. Therefore, the output signal of the VCO locks to the new, albeit incorrect, frequency of the input clock signal, and the lock indicator again reports a valid phase lock status. The reference signal thus clocks the incoming and out-going data at the wrong points resulting in erroneous communication. The input clock signal may also become stuck at the logical value of one or stuck at the value of zero causing the PLL to permanently lose phase lock. Since the PLL cannot lock to a DC signal, the phase lock indicator suspends the system operation. While it is informative to know of the permanent loss of phase lock, the phase lock indicator does nothing toward restoring operation of the system which may remain down until the input clock signal is repaired. In many applications, it is desirable and even imperative that the PLL remain operational even if the primary input clock signal becomes invalid as a reference. As a result, a design has been introduced involving applying redundant input clocks signals to the phase lock loop. In such a design, the validity of the first input clock signal is determined. If the first input clock signal becomes invalid, then there is a switch to utilize the second input clock signal. Such switching is referred to as a "switchover." Unfortunately, such switchovers may occur over a lengthy duration of time (e.g., 156 picoseconds) that results in discontinuities with the interface bit rate thereby causing difficulties in tracking such switchovers.

[0145] Embodiments of the present disclosure improve such technology by implementing a circuit for reducing a time to switch between the redundant clock signals applied to a phase lock loop. A phase frequency detector of the phase lock loop of the circuit detects a shift in the phase of a reference clock signal (one of the redundant input clock signals applied to the phase lock loop) relative to the phase of a feedback clock signal. A feedback clock signal, as used herein, refers to the clock signal outputted by the phase lock loop. The circuit further includes a coarse tuning mechanism configured to delay the feedback clock signal until the feedback clock signal is aligned with a backup reference clock signal (the other redundant input clock signal) in response to

detecting the shift in the phase of the reference clock signal relative to the phase of the feedback clock signal. Such a coarse tuning enables the switchover to occur with a period of the clock frequency. Furthermore, the circuit includes a fine tuning mechanism configured to align the edge of the feedback clock signal with the edge of the backup reference clock signal. In one embodiment, fine tuning is performed by utilizing a series of delay elements to delay the backup reference clock signal at different points in time. In one embodiment, fine tuning is performed by enabling the feedback clock signal to latch data into latches, such as flip-flops, where the data corresponds to an edge of one of the delayed backup reference clock signals. The outputs of such latches are used by an architecture (e.g., XOR logic gates) to convert a thermometer code based on such outputs to decimal logic which is used by a selector to select the appropriate delayed backup reference clock signal, which corresponds to the backup reference clock signal whose edge is aligned with the edge of the feedback clock signal. In one embodiment, such an alignment is performed within the duration of time of a single delay element. As a result, the switchover occurs within the duration of time of a single delay element, such as an inverter (e.g., 19 ps). In this manner, the time duration involved in switching between an invalid input clock signal to a valid input clock signal that is propagated to the phase lock loop is greatly reduced. Furthermore, in this manner, there is an improvement in the technical field involving a phase lock loop.

[0146] The descriptions of the various embodiments of the present disclosure have been presented for purposes of illustration, but are not intended to be exhaustive or limited to the embodiments disclosed. Many modifications and variations will be apparent to those of ordinary skill in the art without departing from the scope and spirit of the described embodiments. The terminology used herein was chosen to best explain the principles of the embodiments, the practical application or technical improvement over technologies found in the marketplace, or to enable others of ordinary skill in the art to understand the embodiments disclosed herein.

Claims

1. A circuit for reducing a time to switch between redundant clock signals applied to a phase lock loop, comprising: said phase lock loop comprising a phase frequency detector configured to detect a shift in a phase of a reference clock signal of two redundant input clock signals applied to said phase lock loop relative to a phase of a feedback clock signal; a coarse tuning mechanism configured to delay said feedback clock signal until said feedback clock signal is aligned with a backup reference clock signal of said two redundant input clock signals; and a fine tuning mechanism configured to align an edge of said feedback clock signal with an edge of said backup reference clock signal of said two redundant input clock signals.
2. The circuit as recited in claim 1, wherein said fine tuning mechanism comprises one or more delay elements configured to delay said backup reference clock signal.
3. The circuit as recited in claim 2, wherein said one or more delay elements comprise one or more inverters.
4. The circuit as recited in claim 1, wherein said fine tuning mechanism comprises one or more latches.
5. The circuit as recited in claim 4, wherein said one or more latches comprise one or more flip-flops.
6. The circuit as recited in claim 1, wherein said fine tuning mechanism comprises an architecture for converting a thermometer code to decimal logic based on outputs of one or more latches.
7. The circuit as recited in claim 6, wherein said architecture comprises a plurality of XOR logic gates.
8. The circuit as recited in claim 6, wherein an output of said architecture is used for selecting one of a plurality of delayed backup reference clock signals whose edge is aligned with said edge of said feedback clock signal.

9. The circuit as recited in claim 8, wherein said selecting is performed by a multiplexer.

10. The circuit as recited in claim 1, wherein said coarse tuning mechanism comprises a shift register or a counter.

11. The circuit as recited in claim 1, wherein said coarse tuning mechanism comprises a divider configured to divide a frequency of said feedback clock signal to match a frequency of said backup reference clock signal.

12. A method for reducing a time to switch between redundant clock signals applied to a phase lock loop, the method comprising: detecting a shift in a phase of a reference clock signal of two redundant input clock signals applied to said phase lock loop relative to a phase of a feedback clock signal; performing a coarse tuning operation corresponding to delaying said feedback clock signal until said feedback clock signal is aligned with a backup reference clock signal of said two redundant input clock signals; and performing a fine tuning operation corresponding to aligning an edge of said feedback clock signal with an edge of said backup reference clock signal.

13. The method as recited in claim 12 further comprising: dividing a frequency of said feedback clock signal to match a frequency of said backup reference clock signal by a divider.

14. The method as recited in claim 13 further comprising: setting said divider to a period of a clock frequency.

15. The method as recited in claim 12 further comprising: outputting said feedback clock signal by a selector in response to detecting said shift in said phase of said reference clock signal relative to said phase of said feedback clock signal thereby causing a phase frequency detector of said phase lock loop to receive said feedback clock signal at both inputs of said phase frequency detector.

16. The method as recited in claim 12, wherein a time duration of said aligning of said edge of said feedback clock signal with said edge of said backup reference clock signal corresponds to a delay of a delay element.

17. The method as recited in claim 16, wherein said delay element comprises an inverter.

18. The method as recited in claim 12 further comprising: enabling said feedback clock signal to latch data into flip-flops, wherein said data corresponds to an edge of one of a plurality of delayed backup reference clock signals.

19. The method as recited in claim 18 further comprising: converting a thermometer code to decimal logic based on outputs of said flip-flops; and selecting one of said plurality of delayed backup reference clock signals using said decimal logic whose edge aligns with said edge of said feedback clock signal.

20. The method as recited in claim 19 further comprising: propagating said aligned backup reference clock signal to said phase lock loop.
